

Functional genetic report · longevity

Exceptional Longevity

Sérgio F · Report date: 02-10-2026

The genes that are most repeated in the studies of centenarians and life expectancy of parents, and polygenic scores of longevity and mortality. Only genes (chip + local imputation + polygenic scores).

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Summary

29

SNPs analyzed

26

in the chip

3

imputed ($DR2 \geq 0.8$)

0

SNP not found in the data

4

Polygenic scores

Exceptional longevity is only ~25% hereditary, but the few genes that replicate in centenarian studies are informative: APOE (the strongest), FOXO3 (the second most replicated), cholesterol genes (CETP, SORT1, LDLR, PCSK9, Lp(a)), 9p21 and inflammation genes. The life expectancy studies of parents in hundreds of thousands of people show that almost all the effect goes through cardiovascular disease, Alzheimer's disease and cancer.

In summary: the genetic basis for getting too far is good. APOE $\epsilon 3/\epsilon 3$ (without $\epsilon 4$), FOXO3 homozygous for the 4 longevity alleles and the 330 longevity score variants in the 92nd percentile. The weaknesses are almost all vascular and inflammatory and are modifiable: 9p21 homozygous risk, CETP high activity (without any of the centenary 3 alleles), IL6 -174 GG and absence of KL-VS.

Favourable

- **APOE $\epsilon 3/\epsilon 3$** (rs429358 TT, rs7412 CC): without $\epsilon 4$, the highest genetic mortality factor.
- **FOXO3 homozygous for the 4 alleles** (rs2802292 GG, rs2764264 CC, rs13217795 CC, rs2802288 AA).
- **Longevity score in the 92nd percentile** (PGS000906, 330 variants).
- **Loci of favourable life expectancy:** SORT1 and LDLR with 1 allele of LDL low each; without Lp(a) high; SH2B3 CC; HFE C282Y absent.
- **Others:** IGF1R A (lower IGF-1), IL10 -1082G and ADIPOQ +276T, all heterozygous.

Unfavourable

- **9p21 homozygous risk** (rs1333049 CC): the locus that shortens parental life expectancy outside APOE and Lp(a).
- **High activity CETP:** without I405V, without TaqIB B2 and without rs3764261 A (the 3 alleles of the centenaries).
- **IL6 -174 GG** (more IL-6; less frequent in centenarians).
- **Without KL-VS** (typical klotho) and without APOE $\epsilon 2$; APOC1 rs4420638 AG (weak effect without $\epsilon 4$).
- **Age of death of the father** (398 variants) in the 30th percentile and male mortality in the 61 (typical); TP53 Arg72/Arg72.

Interpretation: the brain protection and stress response genes (APOE, FOXO3) are on the good side; the arterial protection genes (9p21, CETP) are on the bad side. As almost all the genetics of life expectancy passes through the arteries, keeping LDL/ApoB as low as possible is the intervention that most transforms this profile.

How to read this report

Genetics only. All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 26 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 0 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

Origin and trust: chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with $DR2 \geq 0.8$ (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

Effect and judgment allele: The comparison between the number of Sérgio F copies and the mean of the reference population ($2 \times$ allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

Polygenic scores (PGS): add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage ≥ 80 in a risk score = unfavourable; ≤ 20 = favourable.

Polygenic scores

The longevity score (Tesi 2021, 330 variants, from centenarian studies) is the most informative. The mortality (Meisner 2020) combines scores of diseases; the age of death of the father (Tanigawa 2022) is indirect. In longevity scores, a high percentile is favourable; in mortality, unfavourable.

Trait	Score	Variables used (coverage)	Percentage (reference population)	Judgement
Longevity (330 variants)	PGS000906	312 de 330 (94%)	p92	Favourable
Exceptional longevity (4 variants)	PGS002795	4 de 4 (100%)	p47	Typical / no judgement
Age of death of the father (398 variants)	PGS001393	328 de 398 (82%)	p30	Typical / no judgement
Mortality by all causes, men (4 092 variants)	PGS000319	3 716 de 4 092 (91%)	p61	Typical / no judgement

APOE and TOMM40 region

APOE is the only gene with strong genomic association and consistent with longevity in all major studies. $\epsilon 4$ short (Alzheimer, coronary disease); $\epsilon 2$ is enriched in centenarians. TOMM40 and APOC1 are in the same region.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
APOE $\epsilon 4$ (Cys112Arg)	rs429358	TT	imputation DR2=0.99	C	0	0.15	APOE $\epsilon 4$: the strongest genetic mortality factor in longevity studies (Alzheimer and coronary disease); rare in centenarians. (strong evidence)	Favourable
APOE $\epsilon 2$ (Arg158Cys)	rs7412	CC	chip	T	0	0.07	APOE $\epsilon 2$: the most enriched allele in centenarians (less Alzheimer and lower LDL). (strong evidence)	Typical / no judgement
APOC1/ TOMM40	rs4420638	AG	imputation DR2=1.00	G	1	0.20	Marker of the APOE/APOC1 region (partly linked to $\epsilon 4$): lower longevity. (moderate evidence)	Unfavourable
TOMM40	rs2075650	AA	chip	G	0	0.13	TOMM40 (import of mitochondrial proteins), APOE region: less longevity. (moderate evidence)	Typical / no judgement

Favourable

- APOE $\epsilon 3/\epsilon 3$ (without $\epsilon 4$)
- TOMM40 rs2075650 AA

Unfavourable

- Without $\epsilon 2$
- APOC1 rs4420638 AG (regional marker; $\epsilon 4$ is absent, so the effect is small)

Interpretation: $\epsilon 3/\epsilon 3$ is the reference genotype (~60% of the population). The great advantage is not having $\epsilon 4$.

FOXO3 and the insulin/IGF-1/sirtuins axis

FOXO3 is the transcription factor of stress resistance, autophagy and maintenance of stem cells, blocked by insulin/IGF-1 and mTOR. It is the second most replicated longevity gene.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
FOXO3	rs2802292	GG	chip	G	2	0.43	FOXO3: G allele is the most replicated longevity allele (various peoples); more FOXO3 expression under stress. (strong evidence)	Favourable
FOXO3	rs2764264	CC	chip	C	2	0.37	FOXO3: C allele associated with longevity. (moderate evidence)	Favourable
FOXO3	rs13217795	CC	chip	C	2	0.36	FOXO3: C allele associated with longevity. (moderate evidence)	Favourable
FOXO3	rs2802288	AA	chip	A	2	0.43	FOXO3: allele A associated with longevity. (moderate evidence)	Favourable
IGF1R	rs2229765	GA	chip	A	1	0.43	IGF1R: A allele associated with lower free IGF-1 and longevity in some studies. (weak evidence)	Favourable
SIRT6	rs107251	TC	chip	—	—	—	SIRT6 (DNA repair, heterochromatin): weak association with survival in the elderly. (weak evidence)	Typical / no judgement
SIRT1	rs7895833	AA	chip	—	—	—	SIRT1: promoter polymorphism with no consistent effect on longevity. (weak evidence)	Typical / no judgement

Favourable

- FOXO3: 4 of 4 alleles of longevity in double dose
- IGF1R rs2229765 GA (1 lower allele of IGF-1)

Unfavourable

- None

Interpretation: To be homozygous for the longevity alleles of 4 SNPs of FOXO3 places Sérgio F in a minority group of the reference population, the most associated with reaching 95–100 years. SIRT1 and SIRT6: common polymorphisms without defined effect.

Centenary lipids: CETP, LDL, Lp(a) and triglycerides

In centenarian studies, they have lower CETP and larger lipoprotein particles; parental life expectancy studies highlight SORT1, LDLR, PCSK9 and LPA, all through LDL and Lp(a).

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
CETP I405V	rs5882	AA	chip	G	0	0.33	CETP Val405: less CETP, larger HDL and LDL particles; enriched in centenarians. (moderate evidence)	Typical / no judgement
CETP TaqIB	rs708272	GG	chip	A	0	0.42	CETP B2 (allele A): lower activity CETP and higher HDL. (moderate evidence)	Typical / no judgement
CETP	rs3764261	CC	chip	A	0	0.30	CETP: A allele with higher HDL and less CETP activity. (moderate evidence)	Typical / no judgement
CELSR2/ SORT1	rs12740374	GT	chip	T	1	0.21	SORT1: T allele lowers LDL (~0.16 mmol/L) and increases parental life expectancy (life span locus). (strong evidence)	Favourable
LDLR	rs6511720	GT	chip	T	1	0.11	LDLR: T allele lowers the LDL; locus of parental life expectancy. (strong evidence)	Favourable
PCSK9 R46L	rs11591147	GG	chip	T	0	0.02	PCSK9 R46L LDL ~15% lower for life, less coronary disease. (strong evidence)	Typical / no judgement
ANGPTL4 E40K	rs116843064	GG	chip	A	0	0.03	ANGPTL4 E40K: lower triglycerides and less coronary disease. (strong evidence)	Typical / no judgement
LPA	rs10455872	AA	chip	G	0	0.07	High Lp(a): one of the loci that shorten parental life expectancy. (strong evidence)	Favourable

Favourable

- SORT1 rs12740374 GT and LDLR rs6511720 GT (lower LDL)
- LPA rs10455872 AA (without high Lp(a))

Unfavourable

- CETP rs5882 AA, rs708272 GG and rs3764261 CC (high CETP activity; Genetically lower HDL)
- PCSK9 R46L and ANGPTL4 E40K absent (rare)

Interpretation: High CETP transfers cholesterol from HDL to ApoB particles; the effect is null when LDL/ ApoB is very low. Lp(a) without risk alleles is one of the greatest advantages of this profile.

Other life expectancy loci (9p21, SH2B3, HFE, TP53, ACE)

Loci with effect on the life expectancy of parents (UK Biobank and LifeGen, >1 million parents) and classic genes from centenarian studies.

Polygenic scores of this theme: Longevity (330 variants) (PGS000906): the 92nd percentile **Favourable**; Exceptional longevity (4 variants) (PGS002795): the 47th percentile **Typical / no judgement**; Age of death of the father (398 variants) (PGS001393): the 30th percentile **Typical / no judgement**; Mortality by all causes, men (4 092 variants) (PGS000319): the 61st percentile **Typical / no judgement**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
CDKN2B-AS1 (9p21)	rs1333049	CC	chip	C	2	0.48	9p21: locus of coronary heart disease and shorter parental life expectancy. (strong evidence)	Unfavourable
SH2B3/ATXN2 R262W	rs3184504	CC	chip	T	0	0.46	SH2B3 Trp262: more coronary and autoimmune disease; shorter parental life expectancy. (strong evidence)	Favourable
HFE C282Y	rs1800562	GG	chip	A	0	0.04	HFE C282Y: homozygotes have shorter life expectancy (iron overload). (strong evidence)	Favourable
TP53 Arg72Pro	rs1042522	CC	imputation DR2=0.97	G	0	0.28	TP53 Pro72: less apoptosis and more senescence; it is associated with greater survival after the 85 years despite more cancer. (weak evidence)	Typical / no judgement
ACE I/D (marker)	rs4343	GA	chip	—	—	—	ACE I/D: D allele was associated with French centenarians (not consistently replicated). (weak evidence)	Typical / no judgement

Favourable

- SH2B3 rs3184504 CC
- HFE C282Y absent
- P92 longevity score

Unfavourable

- 9p21 rs1333049 CC
- TP53 Arg72/Arg72 (without Pro72 of the elderly of 85+)

Interpretation: The longevity score (330 variants) is in the 92nd percentile and that of male mortality in the 61: together, typical to favourable. The father's death age score (p30) is typical. The 4 score of exceptional longevity (p47) has little information. ACE I/D: marker without consistent effect.

Klotho, inflammation (IL6, IL10) and adiponectin

Klotho (anti-aging hormone of the kidney), IL-6 (central cytokine of inflammation), IL-10 (anti-inflammatory) and adiponectin (insulin sensitivity; high in centenarians).

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
KL (klotho) KL-VS (F352V)	rs9536314	TT	chip	G	0	0.18	KL-VS: heterozygotes have more circulating klotho, better cognition, less coronary disease and longer longevity. (moderate evidence)	Typical / no judgement
IL6 -174G>C	rs1800795	GG	chip	G	2	0.57	IL6 -174G: more IL-6 production in some studies GG less frequent in men. (weak evidence)	Unfavourable
ADIPOQ -11377C>G	rs266729	CC	chip	G	0	0.28	Adiponectin: G allele is associated with lower adiponectin. (weak evidence)	Typical / no judgement
ADIPOQ +276G>T	rs1501299	GT	chip	T	1	0.27	Adiponectin: T allele is associated with higher adiponectin and less coronary disease. (weak evidence)	Favourable
IL10 -1082A>G	rs1800896	TC	chip	C	1	0.45	IL10 -1082G (high production of anti-inflammatory cytokine): more frequent in centenarians. (weak evidence)	Favourable

Favourable

- IL10 -1082 TC (1 high production allele)
- ADIPOQ +276 GT (1 high adiponectin allele); ADIPOQ -11377 CC

Unfavourable

- IL6 -174 GG
- KL-VS absent (rs9536314 TT): typical klotho

Interpretation: Mixed inflammatory balance: plus genetic IL-6, compensated by a high IL-10 allele. Without KL-VS the klotho is typical; the klotho responds to the exercise and phosphorus of the diet.

Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

Mechanism	Unfavourable	Favourable	Interpretation
Protein aggregation / neuroinflammation	—	APOE ε3/ε3; TOMM40 AA	Favourable (without ε4).
Nutrient Sensing Deregulation	—	FOXO3 4/4 alleles of longevity IGF1R A	Very favourable.
Autophagy / Deregulated Macroautophagy	—	FOXO3 (active autophagy genes)	Favourable.
Exhaustion of Staminary Cells	—	FOXO3 (maintenance of hematopoietic and neural stem)	Favourable.
Vascular Aging	9p21 CC; CETP of high activity	Without high Lp(a); SORT1 and LDLR with low LDL alleles	Main weakness; modified by LDL/ApoB.
Chronic Inflammation / Inflammation	IL6 -174 GG	IL10 -1082G; SH2B3 CC	Mixed.
Intercellular Communication	KL-VS absent	—	Neutral (typical klotho).
Insulin resistance	—	ADIPOQ +276T; FOXO3	Slightly favourable.
Cell Senescence	9p21 (CDKN2A/B) risk; TP53 Arg/Arg	—	9p21 and TP53 Arg72 change the balance between senescence and apoptosis.
Oxidative stress	—	FOXO3; HFE C282Y absent	Favourable.

Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

Topic	Suggestion (not applied)	Genetic basis
LDL/ApoB as low as possible	It is the intervention that most changes this profile: compensates for 9p21 and high CETP. LDL/ApoB: the lower, the better, the lower. <i>suggestion (not applied)</i>	9p21 CC; High CETP; studies of parental life expectancy
Activate FOXO3	Exercise (aerobic + strength), daily feeding window and rapamycin that already takes (mTOR stops FOXO3) potentiate the favourable allele. <i>suggestion (not applied)</i>	FOXO3 4/4
Klotho	Regular aerobic exercise (shoves the klotho); less phosphorus from additives (collar coolers, processed meat, melted cheese). Telmisartan, which already takes, rises the klotho in animal studies. <i>suggestion (not applied)</i>	KL-VS absent
Inflammation (IL-6)	Omega-3 3.2-4 g/day (already taken), regular sleep, exercise, low visceral adiposity and fiber (always). Senolytic pulse with only physetin + quercetin and normal ALT. <i>suggestion (not applied)</i>	IL6 -174 GG
Adiponectin	Exercise, fiber and omega-3 rise to adiponectin; the +276T allele helps. <i>suggestion (not applied)</i>	ADIPOQ
Tracking	Maintaining normal cancer screenings for age (TP53 Arg/Arg and 9p21 change the balance between apoptosis and senescence). <i>suggestion (not applied)</i>	TP53; 9p21
Coordination with the plan	D3 10 000 IU and omega-3 3.2-4 g/day as in the plane. Pausing curcumin, luteolin, apigenin and quercetin while the ALT is high is a suggestion (not applied; the real stack has not changed). No telomerase activators. <i>suggestion (not applied)</i>	V2 plane rules

Limitations

- The longevity is mainly non-genetic (~75%); the centennial alleles have small effects and some were discovered in specific populations.
- APOE was inferred from 2 SNPs (rs429358 imputed with DR2=0.99 and rs7412 of the chip). ACE I/D, exon deletion 3 of GHR and VNTR of SIRT3 are not read by the chip.
- The exceptional longevity score of 4 variants has little information; the age of death of the father covers 82% of the variants.
- The percentiles compare with the reference population (n=633).
- Only genes: does not use blood analysis.

Methods and sources

- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (Beagle 5.5, public reference panel of complete 30x genomes), accepted with $DR2 \geq 0.8$.
- Public positions and alleles: Ensembl REST (only rsIDs). Frequency of alleles calculated locally in the reference population (n=633); the effect allele of each SNP was confirmed by the expected frequency.
- Alleles of effect cured from literature (centenary studies and parental life expectancy).
- Scores: PGS000906 (longevity, Tesi 2021), PGS002795 (longevity > percentile 90, Schoepp 2021), PGS001393 (father's age of death, Tanigawa 2022), PGS000319 (death for all causes in men, Meisner 2020). Percentages vs. reference population (n=633) with exactly the same variants.
- Main references: Deelen 2019 (GWAS longevity, Nat Common); Timers 2019 and 2020 (parental life expectancy, eLife / Nat Common); Willcox 2008 (FOXO3, PNAS); Flachsbart 2009; Barzilai 2003 (CETP, JAMA); Atzmon 2006; Arking 2005 and Dubal 2014 (KL-VS); Bonafè 2001 (IL6 -174); Lio 2003 (IL10); van Heemst 2005 (TP53 Pro72); Sebastiani 2012.
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).